

LAB EXPERIMENT – 6

Vacuum Cleaner Problem using Python — CO3

Aim

To write a Python program to solve the **Vacuum Cleaner Problem** using an intelligent agent.

Procedure

1. Consider two rooms, **A and B**.
2. Each room can either be **Clean** or **Dirty**.
3. The vacuum cleaner can move between the rooms.
4. If the current room is dirty, the vacuum cleaner performs the **Suck** operation.
5. If the current room is clean, the vacuum cleaner moves to the other room.
6. Continue the process until both rooms are clean.
7. Display the actions performed by the vacuum cleaner.

Source Code

main.py	Visualize	Run ▶	Output
<pre>1 def vacuum_cleaner(room_a, room_b, position): 2 rooms = { 3 "A": room_a, 4 "B": room_b 5 } 6 7 print("Initial State:") 8 print("Room A:", rooms["A"]) 9 print("Room B:", rooms["B"]) 10 print("Vacuum Position:", position) 11 print() 12 13 while rooms["A"] == "Dirty" or rooms["B"] == "Dirty": 14 15 if rooms[position] == "Dirty": 16 print("Action: Suck") 17 rooms[position] = "Clean" 18 19 else: 20 if position == "A": 21 position = "B" 22 else: 23 position = "A" 24 25 print("Action: Move to", position) 26 27 print("\nFinal State:") 28 print("Room A:", rooms["A"]) 29 print("Room B:", rooms["B"]) 30 print("Vacuum Position:", position) 31 32 33 # Initial condition 34 vacuum_cleaner("Dirty", "Dirty", "A")</pre>			Initial State: Room A: Dirty Room B: Dirty Vacuum Position: A Action: Suck Action: Move to B Action: Suck Final State: Room A: Clean Room B: Clean Vacuum Position: B

Result

Thus, the **Vacuum Cleaner Problem** was successfully solved using Python by implementing an intelligent agent that cleans both rooms.